EECS 351: Project Topic Finalized Report

1. Finalized Project Topic

Our group has decided to work on the AI-Image Detector project: **AI-Generated Image Detection Using DSP and Machine Learning**

The main idea is to see whether we can distinguish AI-generated images from real photographs by looking at differences in their frequency-domain characteristics. Instead of focusing on one specific frequency, we want to look at how image energy is distributed across different frequency ranges and whether there are any consistent patterns between real and AI-generated images.

We plan to use three DSP techniques: Fast Fourier Transform (FFT), Discrete Cosine Transform (DCT), and Discrete Wavelet Transform (DWT). By comparing the results of these transforms, we hope to find statistical features that can help tell the two types of images apart.

Once we have these features, we will use a simple machine learning classifier, logistic regression classifier or support vector machine (SVM), to test whether they are actually useful for image detection.

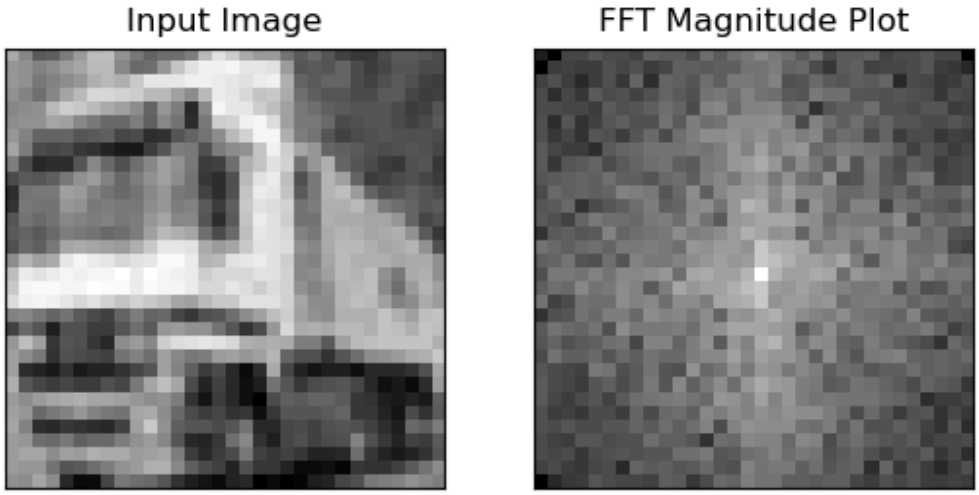
2. Project Approach

Task 1: Dataset Preparation and DSP Analysis

We will use a small, balanced subset of the GenImage dataset, with real images and AI-generated images from one generator. We will first convert the images to grayscale, then apply FFT, DCT, and DWT to each image. We will compare their frequency distributions and extract numerical features, such as the energy in different frequency bands and statistics of the transform coefficients. The goal of this step is to see whether there are noticeable differences between the two image categories and turn those differences into features that we can use for classification.

However, after multiple attempts, we were unable to transform the split ZIP archive into a usable file to extract. So we decided to use the CIFAKE dataset instead. This contains several 32x32 real and generated images. 20 generated images and 20 real images are to be selected to be processed in the following manner: The images will be converted to grayscale, and their 2D FFTs, DWTs, and DCTs would be plotted and saved. We can currently convert the image to grayscale and compute the FFT, but we are still working on computing the DWT and DCT.

For example, this figure was generated with python:



This is the code:

```
import cv2 as cv
import numpy as np
from matplotlib import pyplot as plt
import sys
import pywt

# convert to grayscale
image = cv.imread(sys.argv[1], cv.IMREAD_GRAYSCALE)
assert image is not None, "file could not be read, check with os.path.exists()"
f = np.fft.fft2(image)
fshift = np.fft.fftshift(f)
magnitude_spectrum = 20*np.log(np.abs(fshift))

wavelet = 'db4'
```

```
coeffs = pywt.wavedec(image, wavelet, level=4)

plt.subplot(121),plt.imshow(image, cmap = 'gray')
plt.title('Input Image'), plt.xticks([]), plt.yticks([])
plt.subplot(122),plt.imshow(magnitude_spectrum, cmap = 'gray')
plt.title('FFT Magnitude Plot'), plt.xticks([]), plt.yticks([])
plt.show()
```

Task 2: Image Classification

After extracting the DSP features, we plan to start with Binomial Logistic Regression (we can try this method first and then extend to other methods such as SVM) to classify images as either real or AI-generated.

We will train the model using the training dataset and then test it on images that were not used during training. This will help us see whether the model is learning useful patterns rather than just fitting the training images.

We plan to evaluate the results using accuracy, precision, recall, F1 score, and a confusion matrix. We also want to compare the performance of different feature groups (FFT, DCT, and DWT) to see which ones contribute the most to classification.

Task 3: Other Classification Methods

Besides Logistic Regression, we will look into the other classification methods suggested in the feedback, including SVM, K-Nearest Neighbors (KNN), Naive Bayes, and Decision Trees.

If time allows, we will try these models using the same extracted features and compare their results. This should give us a better idea of whether a more complex classifier actually improves detection performance.

3. DSP vs. Machine Learning

One advantage of traditional DSP techniques is that they allow us to directly examine and understand the frequency characteristics of images. The features are relatively easy to interpret, and we can visualize the differences between real and AI-generated images. However, it may be difficult to distinguish the images using only a few manually chosen thresholds, especially when the differences are subtle. They may also be less computationally expensive than machine learning techniques?

On the other hand, machine learning can help by combining multiple DSP features and learning which patterns are most useful for classification. The downside is that the model may overfit the training data or perform poorly on images generated by a different AI model.

For our project, we think combining DSP with a relatively simple machine learning classifier is a good starting point. It lets us focus on the signal-processing side of the project while still testing whether the features are useful in practice. The following are some other classification methods we investigated:

- KNN
- NaiveBayes
- Classification tree

Traditional DSP techniques can provide an effective and interpretable approach when the classification problem has a clear physical or statistical basis. Because domain knowledge can guide the choice of features, such methods often require less labeled data and computation than complex machine-learning models. However, DSP transforms such as FFT, DCT, and DWT are primarily feature-extraction tools rather than classifiers; the final decision still requires a rule or classifier. Their performance depends heavily on manually selected features and thresholds, and handcrafted features may be sensitive to noise, compression, and preprocessing changes. Consequently, traditional DSP is well suited as an interpretable baseline, but its effectiveness and robustness must be verified experimentally rather than assumed to be universally superior to machine learning.

Using machine learning in this project has several potential advantages. A classifier can learn how to combine multiple DSP-derived features such as FFT band energies, DCT coefficients, and wavelet statistics to make classification decisions, rather than relying on a single manually selected threshold. However, ML also introduces important costs and risks. It requires representative labeled data and a carefully designed train/validation/test split. Otherwise, the model may overfit or exploit dataset-specific shortcuts such as image size, compression, or object category. Its performance may degrade on unseen generators or after compression and resizing, and nonlinear models are often less interpretable than the underlying DSP features. ML also requires feature scaling, hyperparameter tuning, and appropriate evaluation metrics. For these reasons, ML should be treated as a decision layer built on interpretable DSP features, rather than as a replacement for signal analysis.

4. Current Progress and Next Steps

We have finalized our project topic and decided on the overall approach. And we are currently working on Task 1, which involves preparing the GenImage dataset, applying the three DSP transforms, and extracting statistical features. Once this part is completed, we will move on to training and evaluating the Logistic Regression classifier. After that, we plan to explore the other classification methods and compare their performance. We also hope to test how well the model works on images it has not seen before and, if time permits, investigate whether changes such as image compression or resizing affect its performance.

5. Reference

[1] M. Zhu *et al.*, “GenImage: A million-scale benchmark for detecting AI-generated image,” in *Advances in Neural Information Processing Systems*, vol. 36, 2023. [Online]. Available: https://arxiv.org/abs/2306.08571

[2] R. Durall, M. Keuper, and J. Keuper, “Watch your up-convolution: CNN based generative deep neural networks are failing to reproduce spectral distributions,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 7890–7899. doi: 10.1109/CVPR42600.2020.00791.

[3] A. A. Khan, A. A. Laghari, S. A. Inam, S. Ullah, M. Shahzad, and D. Syed, “A survey on multimedia-enabled deepfake detection: State-of-the-art tools and techniques, emerging trends, current challenges & limitations, and future directions,” *Discover Computing*, vol. 28, art. no. 48, 2025. doi: 10.1007/s10791-025-09550-0.

[4] N. A. Bhat and K. J. Giri, “DeepFake detection in images and videos: A survey on models, datasets, and evaluation metrics,” *Multimedia Tools and Applications*, vol. 85, no. 3, art. no. 233, 2026. doi: 10.1007/s11042-026-21365-9.